LEARNER GUIDE

Bearings — describing direction

Constructions, Loci and Bearings · Lesson 6. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

“It's over there” is no use to a rescue helicopter. Sailors, pilots and mountain rescuers describe a direction with a **bearing**: an angle measured **clockwise from North**, always written with **three figures** — so 72° becomes 072°. North is 000°, East 090°, South 180°, West 270°. This lesson teaches you to read and state bearings, to find a **back bearing** (the return direction), and to fix a position from two bearings.

What you are learning to notice

- A **bearing** is measured **clockwise** from **North**.
- It is always written with **three figures** (072°, not 72°).
- The **back bearing** (the reverse direction) differs from the forward bearing by **180°**.
- Two bearings from two places **fix a single position** — where the two bearing lines cross.

Moving through it, one step at a time

To read a bearing of B from A: draw a North line at A, then measure the angle turning **clockwise** from North until you reach the line to B. Write it with three figures. For a back bearing, add 180° if the forward bearing is under 180°, or subtract 180° if it is 180° or more. To fix a position from two bearings, draw a line on each bearing from its station; where they cross is the point.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Reading a bearing

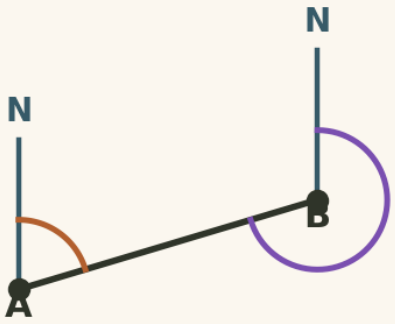
State the bearing of B from A.

1. Draw a North line at A.	North at A
2. Turn clockwise from North to the line AB.	clockwise turn
3. Measure the angle and write three figures.	= 072°

The bearing of B from A is 072° — clockwise from North, written with three figures.

Example 2 — Finding a back bearing

The bearing of B from A is 072°. Find the bearing of A from B.



1. The North lines at A and B are parallel.	parallel Norths
2. The return faces the opposite way — a half turn.	+ 180°
3. 072° is under 180°, so add 180°.	72 + 180

The back bearing (A from B) is 072° + 180° = 252°.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (the four compass directions and measuring an angle in degrees), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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